

Analysis of OSPFv3 in LEO Satellite Networks

Sergio Machado Sánchez
Department of Network Engineering
Universitat Politècnica de Catalunya
Castelldefels, Spain
sergio.machado@upc.edu

Jorge Mata Díaz
Department of Network Engineering
Universitat Politècnica de Catalunya
Barcelona, Spain
jorge.mata@upc.edu

Javier Ozón Górriz
Department of Civil and Environmental Engineering
Universitat Politècnica de Catalunya
Barcelona, Spain
francisco.javier.ozon@upc.edu

Juanjo Alins Delgado
Department of Network Engineering
Universitat Politècnica de Catalunya
Terrassa, Spain
juanjo.alins@upc.edu

Abstract—Low Earth Orbit (LEO) satellite constellations are poised to become key enablers of global connectivity. However, their dynamic topologies, high latency, and intermittent links pose significant challenges for traditional IP routing protocols. This paper evaluates the applicability of OSPFv3 in LEO networks using a novel emulation framework that integrates accurate orbital modeling with full-stack network simulation. Our framework decouples topology generation from network emulation, enabling protocol-independent testing in realistic conditions. We analyze OSPFv3 performance using the Iridium constellation as a case study, assessing protocol convergence, stability, and routing efficiency under worst-case scenarios. Results reveal the limitations of default OSPFv3 configurations in LEO environments and offer insights into potential optimizations for future integration of satellite networks into global IP infrastructures.

Index Terms—LEO satellites, OSPFv3, network emulation, dynamic topology, inter-satellite links, routing protocol evaluation.

I. INTRODUCTION

Low Earth Orbit (LEO) satellite networks are satellite systems that operate at altitudes between 500 and 2,000 kilometers above Earth. Unlike traditional geostationary satellites, LEO satellites orbit the planet more quickly and in greater numbers, enabling low-latency high-speed communication. These networks are increasingly used to provide global broadband coverage, especially in remote or underserved areas, and play a growing role in modern telecommunications infrastructure.

LEO satellite networks are composed of M satellites uniformly distributed across N orbital planes, typically with an inclination below 90° . Each satellite is a cube or a modular unit built from $10 \times 10 \times 10$ cm cubes, with antennas on five faces—including the one oriented toward Earth's center. The Earth-facing antenna provides ground-satellite links (GSLs), while two lateral faces establish intra-plane inter-satellite links (ISLs) with neighboring satellites in the same orbit. These

links are typically stable due to fixed orbital spacing. The other two faces maintain inter-plane ISLs with satellites in adjacent co-rotating planes that share a similar angular phase. These links vary in distance and angle, especially near the poles, and are disabled beyond latitudes of $\pm 60^\circ$ due to geometric and antenna beamwidth limitations.

Figure 1 illustrates a typical LEO satellite constellation. Each vertical arc represents a distinct orbital plane, with satellites connected by solid lines indicating intra-plane ISLs and dotted lines for inter-plane ISLs. At the so-called *seam*, where adjacent orbital planes rotate in opposite directions (counter-rotating), no inter-plane links are possible, resulting in a discontinuity in connectivity. These topological characteristics directly influence routing, latency, and fault tolerance in LEO satellite network design.

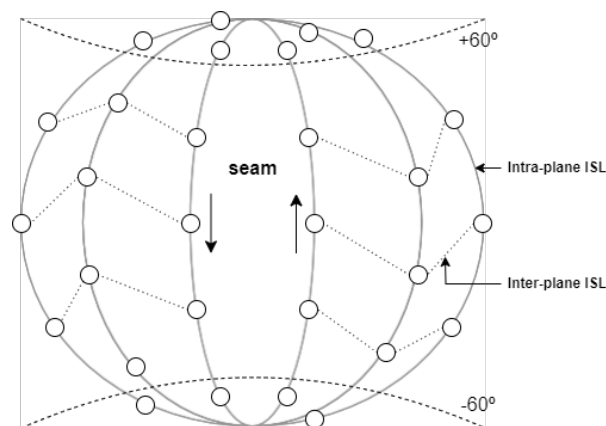


Fig. 1. LEO satellite constellation.

Our research focuses on applying the Low Latency Low Loss Scalable Throughput Internet Service (L4S) architecture [13], originally designed for terrestrial networks, to large-scale LEO satellite communications. The aim is to develop algorithms and protocols that enable seamless integration between LEO networks and ground-based infrastructures, with special

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attention to aeronautical communications. Satellite networks, known for overcoming geographic barriers and offering secure communications, present unique challenges such as high latency, packet loss, and constantly changing topologies, which this work seeks to address.

This work explores the use of the OSPFv3 routing protocol in satellite constellations composed of satellites interconnected via Inter-Satellite Links (ISLs) and linked to ground stations through Ground-to-Satellite Links (GSLs). These constellations create multiple potential communication paths that, combined with high latency and constantly changing topology, pose significant routing challenges [4]. In this context, it becomes essential to conduct an in-depth analysis of OSPFv3, one of the leading candidates for routing in Low Earth Orbit (LEO) satellite constellations. This link-state protocol, widely established in terrestrial infrastructures, is based on maintaining an updated topological database at each node and computing optimal paths using Dijkstra's algorithm. Its design incorporates features that make it a strong candidate for highly dynamic space environments.

Our study analyzes the unmodified OSPFv3 protocol with adjusted configuration parameters using a custom emulation framework. The ultimate goal is to evaluate its viability in enabling reliable communication between aircraft and control centers through global coverage satellite networks.

This paper is organized as follows. Section II reviews related work on routing protocols in satellite networks. Section III presents the emulation framework developed to evaluate the performance of OSPFv3 in LEO satellite constellations. Section IV details the OSPFv3 analysis, describing the emulated scenario and key findings. Finally, Section V draws conclusions and outlines future research directions.

II. RELATED WORK

The design of routing algorithms in Low Earth Orbit (LEO) satellite networks remains a critical research area due to their highly dynamic topology. Frequent link interruptions and topological changes, especially in inter-satellite links (ISLs), introduce significant challenges to the scalability, stability, and efficiency of conventional routing protocols. These difficulties are reflected in the ongoing standardization efforts led by organizations such as 3GPP and the IETF to adapt routing strategies for LEO satellite environments [17].

Researchers have proposed various strategies to address these issues, including adapting distributed protocols originally developed for terrestrial networks and employing centralized approaches, often based on Software-Defined Networking (SDN). Distributed methods offer autonomous management of topological dynamics and benefit from native IP compatibility, facilitating seamless integration with existing ground networks. In contrast, centralized strategies divide the network operation into discrete snapshots where topology is considered static during each interval. Routes are precomputed offline and sent in advance to the network nodes, which apply them synchronously. While effective in predictable environments,

centralized schemes may exhibit reduced resilience to unexpected link failures [17].

Previous evaluations of distributed protocols have discarded the applicability of OLSR (Optimized Link State Routing) in satellite environments. Although designed for ad-hoc networks, OLSR is unsuitable for low node counts and fast link changes, and exhibits limited efficiency when using mechanisms such as multipoint relaying [8]. Similarly, reactive protocols such as AODV have shown limited viability in broadband satellite connectivity, particularly for large-scale LEO constellations. Existing ad hoc approaches that depend on distance vector routing or flooding are generally inadequate for highly mobile environments [11].

Among the distributed protocols studied, OSPF has received significant attention. However, the literature presents contradictory findings regarding its effectiveness. Some studies suggest that minor enhancements are sufficient to adapt OSPF to LEO networks, while others argue that a comprehensive redesign is necessary. Most works proposing modifications to OSPF evaluate their enhancements against the baseline protocol using default terrestrial configurations. This often leads to misleading conclusions, as inappropriate simulations, flawed emulation environments, inadequate parameter tuning, or unsuitable chosen evaluation metrics may fail to accurately reflect the true behavior of LEO networks.

In response to this fragmented research landscape, the present study aims to conduct a systematic and thorough evaluation of OSPFv3 in LEO constellations. The objective is to identify the operational issues and challenges of the protocol and propose adaptations that enable its efficient use in future space-ground integrated infrastructures, where control plane stability is vital for next-generation services. One of the few comprehensive studies in this area is by Wang et al. [8], which highlights the lack of a reliable reference for OSPF performance in satellite systems. Their results indicate that OSPF suffers from significant packet loss and routing overhead in large-scale LEO deployments, making it unsuitable without substantial adaptation. Other comparative studies, such as that by Yan et al. [14], contrast OSPF with the Named-Data Link-State Routing protocol, a link-state algorithm for Information-Centric Networking (ICN). Their findings suggest that OSPF outperforms its counterpart in terms of convergence speed and control overhead.

Several enhancements to OSPF have been proposed to improve its scalability in LEO networks. For instance, Zhang et al. [6] introduce Area-based Satellite Routing, which partitions the constellation into stable inter-area segments to reduce routing updates. Li et al. [15] define clusters within four quadrants and adopt a joint intra- and inter-area routing scheme based on periodic link-state updates.

The hybrid approach, which combines distributed protocols with centralized precomputed routes, has also gained attraction. Fischer et al. [5] were the first to propose Predictable Link-State Routing, which leverages the predictability of satellite motion. Based on this approach, Yan et al. [19] developed SATNET-OSPF, an extension of OSPF that uses cross-layer

techniques for early detection of predicted disconnections and proactive route updates. Wang et al. [7] improve the OSPF protocol incorporating orbital area division, preloaded link-state databases, and refined neighbor state machines. A recent contribution by Wang et al. [18] enhances this approach with Deep Q-Network-based dynamic adaptations to congestion.

Geolocation-based forwarding strategies offer another line of research. Li et al. [12] present SER, a routing algorithm that computes shortest paths using only orbit and satellite IDs, thereby avoiding link-state flooding and reducing intermediate node overhead. Wang et al. [20] introduced MPLF, a distributed, convergence-free architecture that incorporates satellite orientation and location into next-hop decisions. Yang et al. [16] proposed IFAR, which combines geolocation-aware forwarding with linear programming to match satellites with ground stations while taking into account capacity constraints. Similarly, OPSPF [9] calculates routing tables based on satellite positions in real time and activates on-demand routing only upon detection of unexpected topological changes.

Our work makes a substantial contribution to the re-assessment of the OSPFv3 protocol in a realistic LEO satellite environment. By implementing an innovative emulation framework that combines accurate orbital modeling with comprehensive network simulation, this study establishes a foundational knowledge base and provides valuable insights for the future design of routing protocols in such dynamic systems. The evaluation results demonstrate that OSPFv3 is capable of maintaining robust connectivity and achieving low convergence latency, even under continuous satellite mobility. These findings align with and support the recent guidelines outlined in RFC 9717 [21], which presents a scalable routing architecture for satellite networks based on existing IETF link state routing protocols, enhanced by awareness of scheduled link connectivity changes. This RFC recommends IS-IS as the preferred protocol due to its inherent flexibility in area definition.

To address predictable topological variations, RFC 9717 advocates the proactive assignment of high metrics to links expected to disconnect. This approach enables traffic redirection toward alternative routes before link degradation occurs, facilitating a more secure and coordinated transition compared to abrupt link removal. By avoiding problems such as incomplete state propagation and timing inconsistencies, the strategy ensures network convergence ahead of the actual topological event. This proactive metric assignment mechanism is one of the main conclusions confirmed by our findings.

III. EMULATION FRAMEWORK

This section explains the design and implementation of the LEO satellite network emulation framework, on which we will perform the analysis of OSPFv3 as an IP routing protocol. The design has been made in such a way that it is routing protocol independent, thus allowing us to evaluate different routing solutions, not only OSPFv3.

The framework is structured into two main components. The first focuses on topology generation, producing a sequence

of time-evolving graphs that accurately represent the dynamic behavior of the satellite constellation over time. The second component is responsible for the emulation of network behavior using a routing protocol. In this study, OSPFv3 is used to evaluate its performance in the context of highly dynamic LEO satellite networks.

A. Dynamic Topology Modeling

This subsection presents the module in charge of generating the dynamic topology of the satellite network. It relies on a modified version of the Hypatia orbital propagator [10] to compute satellite positions, active inter-satellite links (ISLs), and visibility with ground stations over time. The original `satgenpy` module was extended to calculate ISL and Ground Station Link (GSL) parameters, determine satellite-ground station visibility, and generate a sequence of GML files describing the network's temporal evolution.

In the orbital propagator, three types of input data are used in the emulator: the configuration of the satellite constellation and its active inter-satellite links, and the set of satellites within communication range of each ground station along with their respective distances. Both of these are provided for each time interval to capture the temporal evolution of the network topology. The third input is the geographical location of the ground stations, which remains static. This information is processed to generate a single JSON file per snapshot, describing the resulting LEO network topology.

Each generated JSON file represents the network topology at a given snapshot through two main components. One component defines a set of labeled virtual networks, each corresponding to an inter-satellite or ground-to-satellite link, using default configuration parameters. The other component describes all nodes in the network, including satellites and ground stations, each identified by a label derived from the input data. For each node, a set of configuration parameters is provided, including those related to the routing protocol, which in this case is OSPFv3.

The generation of each snapshot involves three main stages. First, the spatial configuration of the constellation is extracted, including satellite positions and ISLs with their associated delays. Then, based on the relative positions of satellites and ground stations, the most suitable connections are established using a hard handover criterion. Finally, all this information is combined into a structured JSON description that captures the network state at that instant, serving as the basis for subsequent emulation stages.

B. Network Emulation

The network emulation relies on Linux bridges to model both ISLs and GSLs, enabling the representation of network connectivity and delay in a realistic manner per snapshot. Each network node, whether it is a satellite, a ground station or a traffic sink, is instantiated as a Linux container using Incus, providing an isolated and lightweight environment that supports full network stack emulation. This combination of

containers and virtual networking allows precise control over the topology and its dynamics during the emulation.

The network emulation block is responsible for recreating the dynamic behavior of the satellite network over time. It operates in two main phases. The first phase involves initializing the emulation environment by deploying containers that simulate satellites, ground stations, and the network links between them. These links include both inter-satellite and satellite-to-ground connections, and are configured to reflect realistic transmission delays based on their physical distances.

Once the initial topology is built, the second phase introduces time-based events that reflect the evolution of the network. These events simulate changes in connectivity, such as the appearance or disappearance of links between nodes, or updates to the delay associated with existing links. Events are scheduled using a dedicated event management library, which allows the emulation to reproduce the time-varying nature of the topology throughout the simulation.

Two main types of events are considered: those affecting ISLs and those involving ground station connectivity. For ISLs, events include the activation or deactivation of links and updates to their latency. For ground stations, events reflect the changing visibility of satellites as they move in and out of range. Each event is scheduled to occur at a specific moment during the emulation, and includes the relevant parameters for applying the changes dynamically.

The output of this emulation process consists of detailed packet-level traces. These include routing protocol messages, which allow for the analysis of routing dynamics over time. Additionally, performance metrics such as packet loss, delay, and jitter can be extracted, providing a realistic characterization of how the satellite network behaves under dynamic conditions.

IV. OSPFv3 ANALYSIS IN LEO SATELLITE NETWORKS

In this section, we analyze the performance of OSPFv3 within a LEO satellite constellation. Given the highly dynamic topology and stringent timing requirements of LEO constellations, evaluating OSPFv3's scalability and convergence characteristics in this environment is critical. The Iridium constellation [1], with its global coverage and inter-satellite links, provides an ideal case study to understand the challenges and potential adaptations required for OSPFv3 to operate efficiently in satellite-based networks.

Fig. 2 shows the scenario emulated and analyzed in this section. All tests generate traffic from a single satellite located at the equator over a full orbit of the Iridium constellation. This satellite features six interfaces: two for intra-plane ISLs, one for an inter-plane ISL, one unused due to its position near the seam between planes, one loopback interface for OSPFv3 router identification assignment, and one connecting to the aircraft traffic generator that communicates via ground stations. The analyzed network comprises 66 Iridium satellites and 4 ground stations (in Tempe, Fairbanks, Svalbard, and Punta Arena), which relay data through a terrestrial gateway

to multiple airspace control centers (sinks). This setup represents a worst-case scenario, as satellites do not constantly transmit traffic and aircraft maintain intermittent connections to different satellites during flight.

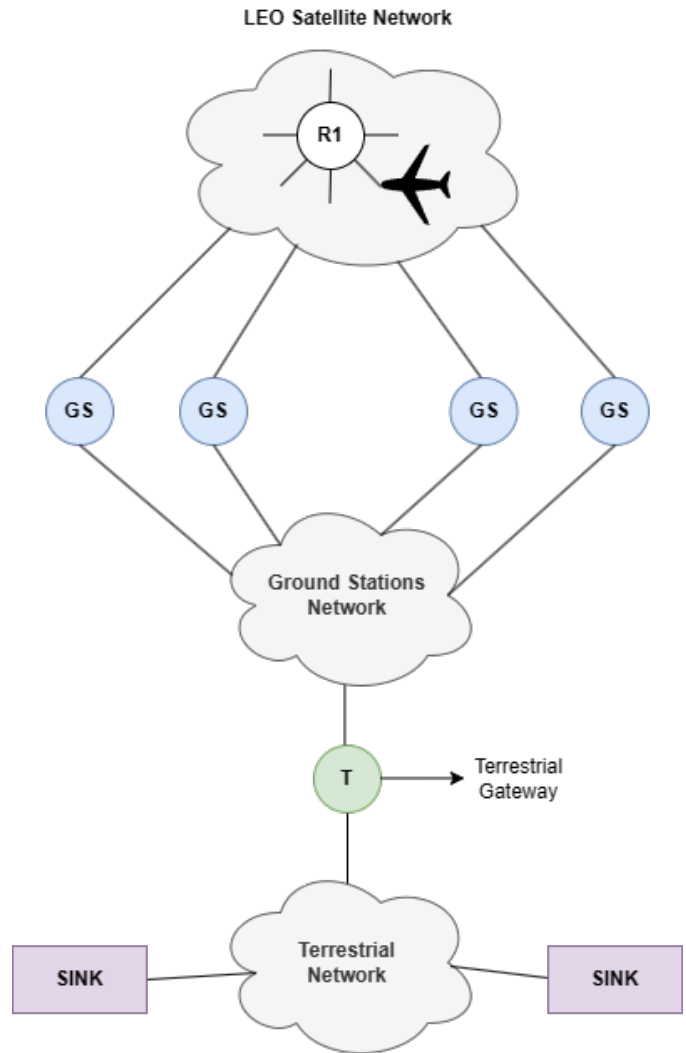


Fig. 2. Schema of the emulation.

A. OSPFv3

OSPFv3 [3], the version of OSPF designed for IPv6, was published in 1999 and updated in 2008, as specified in RFC 5340. It retains the core mechanisms of OSPFv2, such as the flooding algorithm for information distribution and the use of Dijkstra's shortest path algorithm for route calculation. However, OSPFv3 introduces important changes to support IPv6, including modifications to address formats and router identifiers, and the removal of built-in authentication due to IPv6 security features. The protocol organizes routers into areas, but in highly dynamic networks like LEO satellite constellations, the concept of stable areas is challenging to apply.

Routers in OSPFv3 exchange five types of messages directly over the IP layer to build and synchronize their link-state databases, ensuring all routers within the same area have a consistent view of the network topology. Hello messages maintain neighbor relationships, while Link State Advertisements (LSAs) carry detailed network information using a reliable flooding mechanism. This synchronization is essential to compute loop-free, optimal routes. Of the several LSA types defined, this work focuses on Router LSAs, Link LSAs, and Intra-Area Prefix LSAs, which respectively describe router interfaces, neighbor link-local addresses, and IPv6 prefixes within an area.

Despite its strengths, OSPFv3 faces scalability challenges in large satellite networks. The volume of LSAs generated can be very high, increasing control message overhead. Additionally, the computational effort for route calculation grows exponentially with the number of nodes, limiting the protocol's responsiveness to frequent topology changes typical in LEO satellite networks.

The OSPFv3 configuration for satellites includes assigning an IPv4 address from the 10.0.0.X/32 range to the loopback interface, which serves as the router identifier in OSPFv3, and an IPv6 address from the 2002::X/128 range, where X is an 8-bit identifier for the satellite. For ground stations, the IPv4 address is assigned from the 10.X.255.255/32 range, while all ground stations share the IPv6 anycast address 2002::ffff/128. Anycast is used to enable routing toward any available ground station, with the protocol automatically selecting the nearest one. Interfaces connected to ISLs and GSLs are configured as point-to-point links, which prevents the election of designated routers and avoids the generation of Network LSAs. Additionally, the Hello and Dead intervals are reduced to one and two seconds respectively, allowing for rapid neighbor discovery and failure detection in OSPFv3.

B. Results for OSPFv3

The study was carried out by performing two-hour emulations, which exceed one full orbit of the Iridium constellation (100 minutes), using different snapshot intervals. Specifically, analyses were performed with 5-second and 20-second intervals under two distinct transmission configurations:

The results presented encompass the following analyses:

- Distribution of convergence times of OSPFv3 Router LSAs.
- Histogram-based quantification of the hop count distribution between satellites and their nearest ground stations.

Fig. 3 illustrates the satellite ground track over the two-hour emulation period with a snapshot interval of 5 seconds. The satellite initiates its trajectory at the equator above northern Brazil (latitude 0°) and concludes over northern Canada. Variations in the satellite's trajectory directly impact the number of hops required to reach the ground stations, as the inter-satellite and satellite-to-ground distances change dynamically throughout the orbit.

Fig. 4 depicts the distribution of convergence times for OSPFv3 Router LSAs corresponding to the specified snapshot

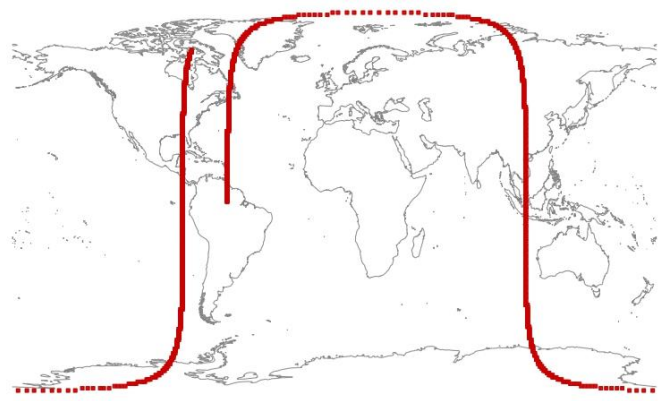


Fig. 3. Satellite ground track, $\Delta t = 5$ s.

interval. The data reveal that the most frequent convergence time, which represents the latency for routers to propagate and assimilate topological changes, lies within the 0 to 1 second interval. The calculated average convergence time across all observed events is 2.1842 seconds.

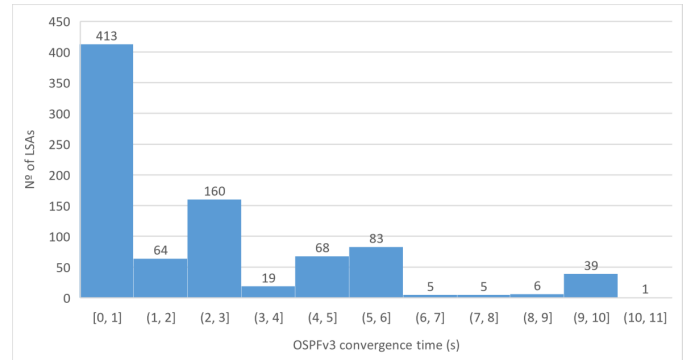


Fig. 4. OSPFv3 convergence time histogram, $\Delta t = 5$ s.

Fig. 5 illustrates the distribution of the number of hops required by the source satellite, defined as the satellite simulating a connected aircraft, to reach the nearest ground station according to the OSPFv3 cost metric for each snapshot. The hop count distribution is relatively uniform with most observations concentrated at distances of three, four and five hops, indicating a stable multi-hop routing pattern within the network topology.

Fig. 6 depicts the satellite ground track over the two-hour emulation period. The snapshots captured at 5-second intervals correspond to the same temporal window, resulting in an identical satellite trajectory, beginning in northern Brazil and concluding in northern Canada. Notably, fewer data points are present, particularly at latitudes beyond 60° North and below -60° South, which is attributed to the increased spacing between consecutive snapshots.

Fig. 7 presents the convergence time distribution of OSPFv3 Router LSAs for this snapshot interval configuration. As observed, the most frequent convergence time once again lies within the 0 to 1 second range, indicating a rapid propagation

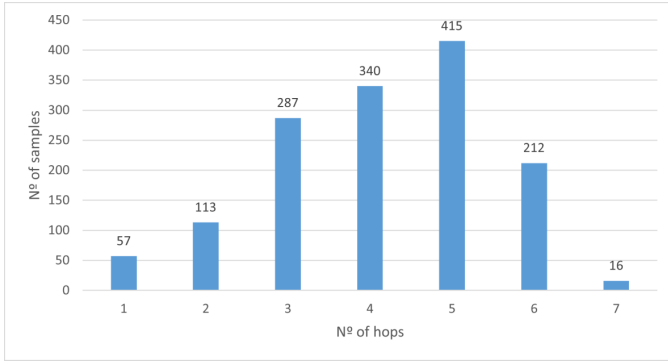


Fig. 5. Number of hops histogram, $\Delta t = 5$ s.

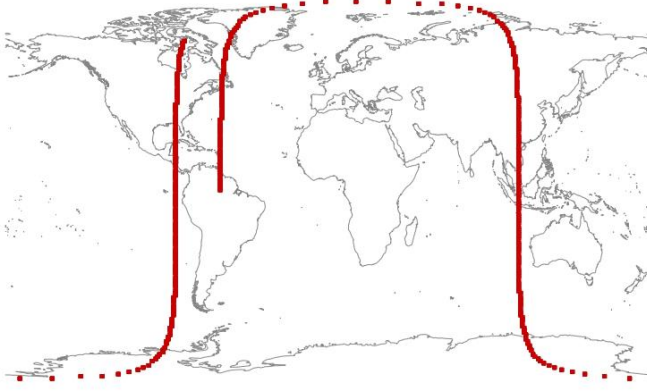


Fig. 6. Satellite ground track, $\Delta t = 20$ s.

of topology changes across the network. The average convergence time measured is 2.3529 seconds.

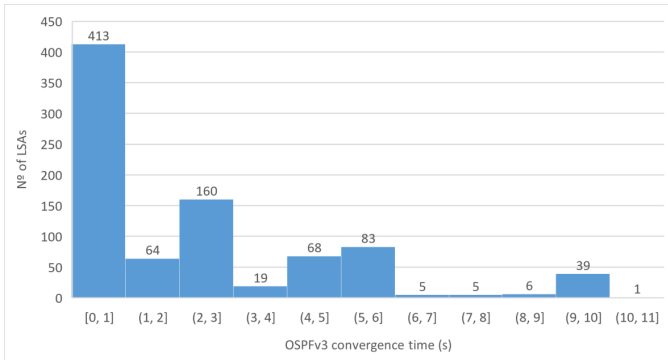


Fig. 7. OSPFv3 convergence time histogram, $\Delta t = 20$ s.

Fig. 8 shows the number of hops it takes source satellite to reach the nearest ground station in each snapshot. With this spacing between snapshots, columns of 3, 4 and 5 hops are dominant.

The results highlight the robustness of OSPFv3 in highly dynamic LEO satellite networks. In both configurations with snapshot intervals of five and twenty seconds, the most frequent convergence time for Router LSAs falls within the zero to one second range, indicating rapid dissemination

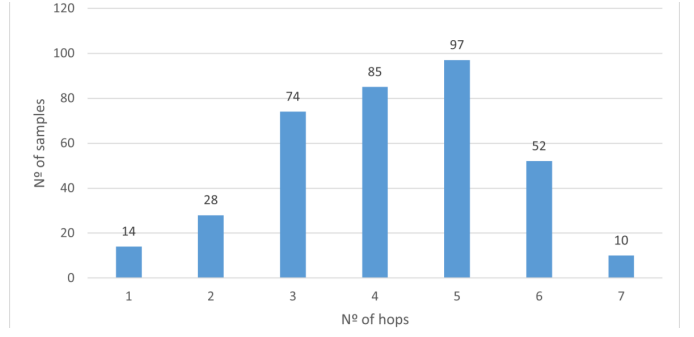


Fig. 8. Number of hops histogram, $\Delta t = 20$ s.

of topology changes throughout the network. The average convergence times, 2.1842 seconds for the five-second interval and 2.3529 seconds for the twenty-second interval, confirm that OSPFv3 maintains low convergence delays even under continuous satellite mobility. The slightly higher average in the latter case may be attributed to the coarser temporal granularity, which can mask finer grained routing dynamics and lead to larger observed transitions between states.

The distribution of hop counts required for a satellite to reach the nearest ground station remains consistent across both scenarios, with 3, 4, and 5-hop paths being the most common. This suggests that the underlying inter-satellite topology provides sufficient redundancy and stability to maintain efficient multi-hop routing. Furthermore, the satellite ground tracks reveal that coverage is more sparse at latitudes beyond $\pm 60^\circ$, where fewer observations are recorded due to increased snapshot spacing. These regions may therefore experience reduced visibility or fewer available paths to ground stations. Overall, the findings indicate that OSPFv3, when tuned appropriately, is capable of delivering fast and stable routing performance in LEO constellations, even as link states change rapidly due to orbital motion.

C. Traffic Performance Evaluation

In this section, we present the end-to-end performance results for the LEO satellite network using OSPFv3 as the routing protocol, with a snapshot interval of $\Delta t = 5$ s. The analysis focuses on key traffic metrics including delay, jitter, throughput, and packet loss. These metrics provide insight into the quality of service experienced by data flows traversing the dynamically changing satellite topology. Traffic generation was carried out using the Distributed Internet Traffic Generator (D-ITG) [2], enabling fine-grained control and measurement of network performance under realistic transmission patterns.

Table I summarizes the key performance metrics collected during the two-hour traffic generation with a snapshot interval of 5 seconds. The network transmitted a total of 357,552 packets at an average rate of approximately 49.66 packets per second. The reception side successfully received 351,769 packets, indicating a packet loss rate of about 1.62%, which reflects moderate network reliability under the tested conditions.

TABLE I
TRAFFIC PERFORMANCE METRICS, $\Delta t = 5$ s

Total time	7200 s
Transmission Metrics	
Total packets sent	357,552 pkt
Average packet rate	49.66 pkt/s
Reception Metrics	
Total packets received	351,769 pkt
Minimum delay	4.186 ms
Maximum delay	131.54 ms
Average delay	46.148 ms
Average jitter	359 μ s
Delay standard deviation	18.221 ms
Average packet rate	48.8568 pkt/s
Packets dropped	5,783 (1.62%)

The observed delays ranged from a minimum of 4.186 ms up to a maximum of 131.54 ms, with an average end-to-end delay of 46.148 ms. The delay jitter was low, with an average of 359 microseconds and a standard deviation of 18.221 ms, suggesting relatively stable latency fluctuations during the emulation. The received packet rate closely matches the transmission rate, confirming consistent throughput throughout the experiment.

Overall, these results demonstrate that the LEO satellite network with OSPF routing achieves satisfactory packet delivery and maintains latency and jitter within reasonable bounds for this configuration, supporting its viability for delay-sensitive traffic.

Fig. 9 presents a comparative analysis of the packet delay and the number of hops observed at each time step throughout the entire emulation period. The two curves exhibit a strong correlation, indicating that an increase in the number of hops corresponds to a proportional increase in packet delay. Notably, the transient peaks lasting approximately 5 seconds are attributed to the short interval between snapshots, which may introduce inaccuracies in path selection during the generation of snapshot files. Additionally, it is important to highlight that packets registering zero delay correspond to dropped packets. An illustrative example of this phenomenon is observed between seconds 400 and 500.

Fig. 10 presents the histogram of packet delays for the received traffic. The distribution exhibits notable concentrations within the intervals (15, 20] ms and (70, 75] ms, with dominant peaks observed in the (30, 35] ms and (40, 45] ms ranges. According to the D-ITG metrics summarized in Table I, the average packet delay under the current scenario is measured at 46.148 ms.

Fig. 11 illustrates the evolution of the packet loss rate over time, alongside the corresponding satellite latitude, in order to assess potential correlations between geographical position and network reliability. A total of seven instances of complete packet loss (100%) are observed. The first occurs at $t = 5$ s and is attributed to the initialization of GSL events, during which links are momentarily dropped and re-established at each ground station. A prolonged loss is detected between 400 s and 500 s, which appears unrelated to network behavior

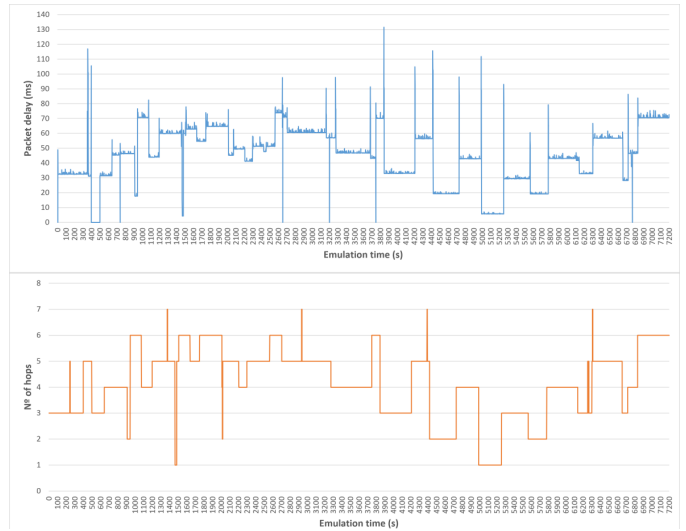


Fig. 9. Packet delay and number of hops, $\Delta t = 5$ s.

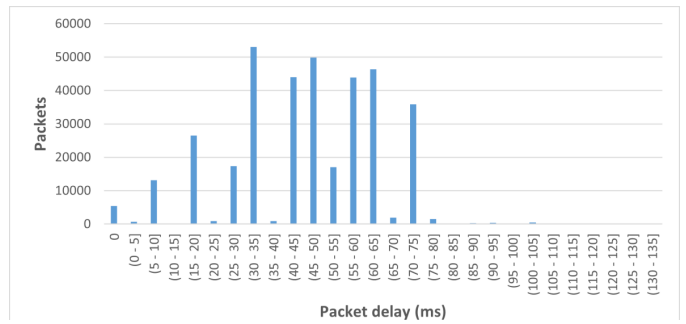


Fig. 10. Packet delay histogram, $\Delta t = 5$ s.

and is instead attributed to internal limitations in D-ITG performance during that interval. Additional losses at 740 s and 6770 s coincide with the satellite crossing the 60° latitude, where the inter-plane ISL drops as part of the network's dynamic topology. Finally, packet losses at 2655 s, 3205 s, and 3750 s occur near the equator, a region characterized by a greater number of hops between the satellite and the ground stations.

The end to end performance evaluation of the LEO satellite network under OSPFv3 routing and a snapshot interval of $\Delta t = 5$ s reveals that the system is capable of maintaining acceptable quality of service metrics in a dynamic orbital context. The average delay and jitter remain within tolerable limits for many delay sensitive applications, and the overall packet delivery rate is consistent with moderate network reliability, as evidenced by a loss rate of 1.62 percent. Correlation analyses indicate that packet delay closely follows the number of hops, confirming the sensitivity of end to end latency to topological variations. Furthermore, the temporal occurrence of packet losses linked to events such as inter satellite link drops, ground station handovers and simulation tool limitations highlights critical periods where routing adaptability or protocol resilience may be challenged. These insights

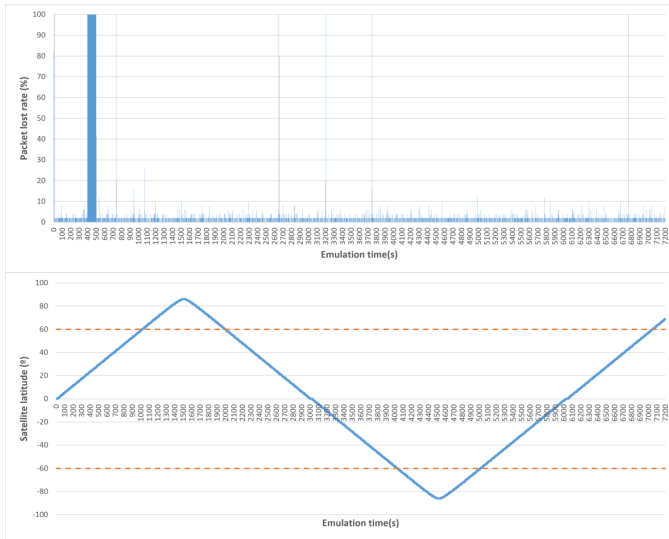


Fig. 11. Packet loss rate and satellite orbit latitudes, $\Delta t = 5$ s.

underscore the importance of robust routing strategies and adaptive mechanisms to mitigate the inherent variability of LEO constellations, especially in regions of high mobility or frequent topological changes.

V. CONCLUSIONS AND FUTURE WORK

This work presents a comprehensive study on the applicability of OSPFv3 in LEO satellite networks, supported by a custom emulation framework that combines realistic orbital dynamics with full-featured network stack simulation. Our results demonstrate that although OSPFv3 can maintain connectivity under dynamic LEO conditions, its default behavior and timing parameters are not well suited for highly time-variant topologies.

The proposed framework represents a significant advancement in the evaluation of routing protocols for satellite communications. By decoupling the topology generation from protocol operation and emulating realistic ground and inter-satellite interactions, it enables precise analysis of protocol performance in complex scenarios. This contribution lays the groundwork for future developments in adaptive routing strategies and highlights the necessity of protocol enhancements to support next-generation space-based Internet services.

Building upon the satellite network emulation framework developed in this study, future work will focus on the evaluation of additional routing algorithms. In particular, we plan to assess not only traditional approaches but also routing strategies specifically designed to exploit the predictable temporal evolution of satellite network topologies. Given the deterministic nature of orbital mechanics, such time-aware algorithms may offer significant improvements in performance and reliability compared to static or purely reactive solutions.

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